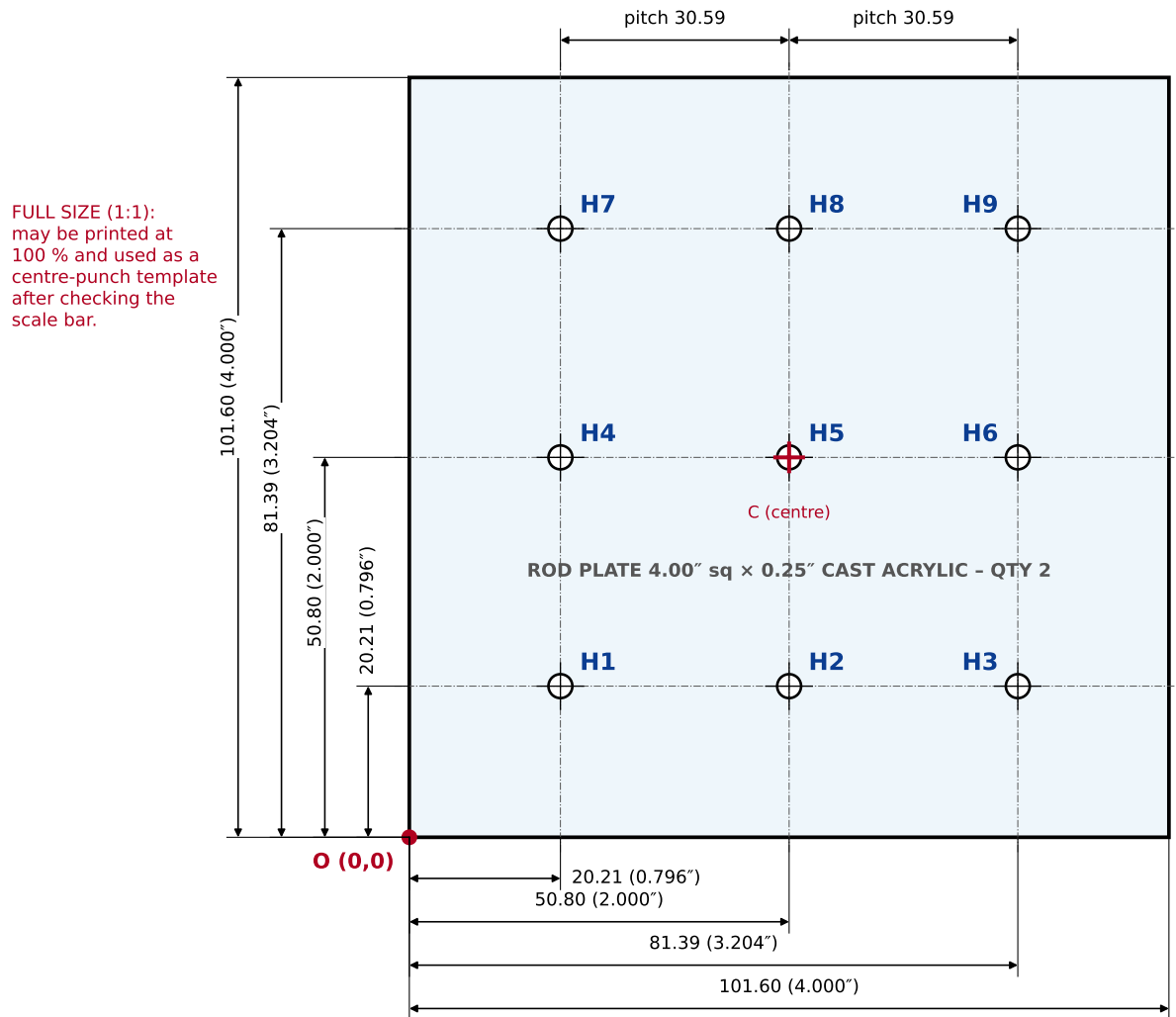


Top view, scale 1:1. 3 × 3 grid, pitch $\lambda/4 = 30.59$ mm. 9 holes $\varnothing 3.175$ mm REAM THRU. Origin O = lower-left corner.

Top view, scale 1:1. 3 × 3 grid, pitch $\lambda/4 = 30.59$ mm. 9 holes $\varnothing 3.175$ mm REAM THRU. Origin O = lower-left corner.



FULL SIZE (1:1):
may be printed at
100 % and used as a
centre-punch template
after checking the
scale bar.

All dimensions mm (inch). Baseline dims from O. 1:1 – print at 100 %, check scale bar, then may be used as a punch template.

2.45 GHz band · HYP (scaled; Dan's extension)

2.45 GHz band · HYP (scaled; Dan's extension)

DRAFT - for Dan's review

Experiment	2.45 GHz oven band (Dan's extension)
Frequency	2450.0 MHz (2.45 GHz)
Wavelength	$\lambda = c/f = 122.36 \text{ mm}$
Rod length	61.18 ± 0.2 mm (2.409")
	$\lambda/2 = 61.18 \text{ mm}$ (free space)
Rod spacing	$\lambda/4 = 30.59 \text{ mm}$ (1.2044")
Rod dia	1/8" = 3.175 mm (brass, then W1)
Hole	Ø3.175 mm REAM THRU, +0.013/-0
	size to MEASURED rod (see sheet 2)
Hole position	±0.13 mm (±0.005")
Material	CAST acrylic, 0.25" (6.35 mm)
Rod plate	4.00 × 4.00" (101.6 mm sq), QTY 2
Scale	TOP VIEW 1:1 on letter (print 100%)
Date / by	2026-09-28 · Grok Bot for Dan Britton
Sheet	1 of 3 · DRILL PATTERN (TOP VIEW)

Hole	X mm	Y mm	X in	Y in
H1	20.21	20.21	0.7956	0.7956
H2	50.80	20.21	2.0000	0.7956
H3	81.39	20.21	3.2044	0.7956
H4	20.21	50.80	0.7956	2.0000
H5	50.80	50.80	2.0000	2.0000
H6	81.39	50.80	3.2044	2.0000
H7	20.21	81.39	0.7956	3.2044
H8	50.80	81.39	2.0000	3.2044
H9	81.39	81.39	3.2044	3.2044

Origin O = lower-left corner. Hole Ø3.175 THRU.

Centre-origin table: sheet 2.

QTY 2 plates, identical drilling.



check: bar must measure 50 mm on paper

Position tol ± 0.13 mm (± 0.005 ").

SHEET 3 - CUT LIST, ASSEMBLY, NOTES - 2.45 GHz band

Part	Qty	Size (in)	Size (mm)	Operation
Rod plate	2	4 × 4"	102 × 102	drill/ream 9 holes (sheet 1)
H-array base	1	4 × 4"	102 × 102	dado 0.25" +0.004 wide × 0.125" deep, centred
V-array legs	2	4 × 2.125"	102 × 54	square ends, no holes
Fit-test coupon	1	3 × 3"	76 × 76	3 trial holes: reamer sizes

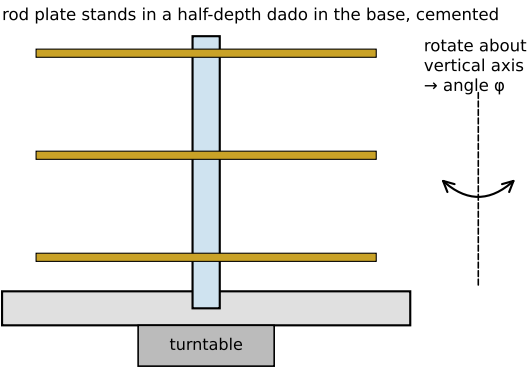
All parts CAST acrylic 0.25" (6.35 mm), clear. Net area 74.0 in²; +20 % margin 88.8 in²; acrylic mass ≈ 0.36 kg.

BUY: 1 sheet 12 × 12" × 0.25" cast acrylic (fits by guillotine nesting, 1/8" kerf) - or order the parts cut to size.

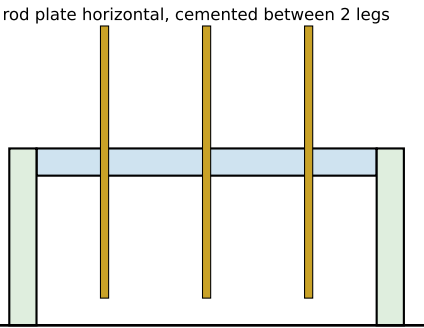
V-array leg height 2.125" = rod protrusion 27.4 mm + 25 mm clearance to the stand (IDEA).

Join: Weld-On 16 (or 4) solvent cement; anneal machined parts before cementing (acrylic makers' advice). NO metal fasteners near the rods.

H ARRAY (rods horizontal, rotatable)



V ARRAY (rods vertical, fixed)



Keep steel (turntable bearing, screws) ≥ λ/(2π) = 20 mm from the nearest rod: add a wood/plastic riser under the base if needed (IDEA).

Stand height: centre rod H5 on the TX-RX ball axis. Height is an OPEN item (depends on ball mounts).

- FACT** Monstein & Wesley, EPL 59(4) 514 (2002) p.516: 'cubic array of 9 wires of length λ/2 (34.6 cm)' as polarizer-analyzer at 433.59 MHz; brass rods; Fig. 2 photo: rods pass through ONE board at mid-length; 'Two such arrays with wires perpendicular to the direction of propagation and to each other'. Rod spacing, rod diameter and board size are NOT stated. Ball antennas 6 cm dia.
- FACT** Butterworth et al., J. Sci. Explor. 27(1) 13-23 (2013) p.16 (446 MHz rebuild): '3 × 3 arrays, a half-wavelength long and a quarter-wave apart, one horizontal and one vertical. The horizontal polarizer can be rotated.' → λ/4 pitch comes from Butterworth.
- HYP** Frequency/scaling: M&W's λ/2 rods and Butterworth's λ/4 pitch scaled to this frequency (Dan's extension; not in either paper).
- HYP** 'Cubic' read as: 3 × 3 at λ/4 spans λ/2, equal to the rod length, so the array fills a λ/2 cube.
- IDEA** Acrylic replaces the earlier HDPE fixtures (rod-fixtures rev A) as requested. Single plate at rod mid-length follows M&W Fig. 2.
- IDEA** Plate t/L = 0.104: plastic covers a small part of each rod. Rods electrically FLOATING, no wires, as in M&W.
- IDEA** 1/8" rods: stock on hand/ordered (brass + W1), same holes for both; half-rod self-weight sag < 0.01 mm (brass).
- NOTE** Rod material enters M&W's model only through σ (eq. 7); geometry is identical for brass and W1 rods.

SLW HUB · ACRYLIC ROD-HOLDING JIG

2.45 GHz band · HYP (scaled; Dan's extension)

DRAFT - for Dan's review

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Hole position	±0.13 mm (±0.005")
Material	CAST acrylic, 0.25" (6.35 mm)
Rod plate	4.00 × 4.00" (101.6 mm sq), QTY 2
Scale	sketches NOT to scale
Date / by	2026-09-28 · Grok Bot for Dan Britton
Sheet	3 of 3 · CUT LIST, ASSEMBLY, NOTES

ASSEMBLY ORDER

1. Cut & drill coupon; pick reamer that gives a snug push fit by hand.
2. Drill/ream both rod plates together (stacked, pinned) so patterns match.
3. Anneal, then cement H plate into base dado; cement V plate between legs.
4. Push rods in, centre them (equal 27.4 mm each side), add O-rings.
5. Brass first; later swap to W1 rods (same holes, same geometry).

Files: numbers.json, src/*.py
/workspace/sim-lab/rod-jig/